Total No. of Pages **2** Roll No.....

VI SEMESTER

B.Tech

MID TERM EXAMINATION ***March-2024***

COURSE CODE CO304 **COURSE TITLE** Artificial Intelligence

Time: 1:30 Hours ***Max. Marks: 25***

Note : All questions are compulsory.
Assume suitable missing data, if any.

Q.1 You are given two jugs, a 4-gallon one and a 3-gallon one, a pump which has unlimited water which you can use to fill the jug, and the ground on which water may be poured. Neither jug has any measuring markings on it. How can you get exactly 2 gallons of water in the 4-gallon jug?

i. Formulate this as an AI problem.

ii. Represent the entire possible state space as a set of rules.

iii. Come up with a possible solution of the problem, specifying which rule is applicable for each state transition.

iv. Represent the solution as a breadth first and depth first search procedure.

v. Discuss the time and space complexity as well as the optimality.

[1.5x5=7.5][CO1]

Q.2 Consider the graph given in Figure 1. The numbers written on edges represent the distance between the nodes. The numbers written on nodes represent the heuristic value. Find the most cost-effective path to reach from start state A to final state J using A* Algorithm, showing at each step the values of f, g, and h.

[5][CO3]

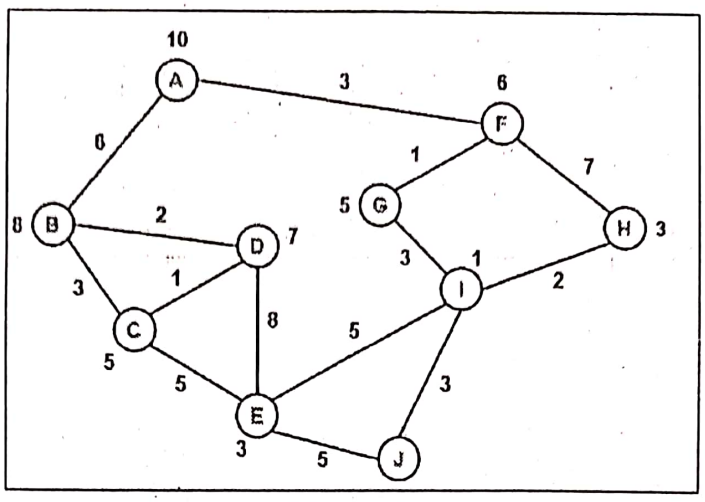

Figure 1

Q.3 For the Figure 2, apply Alpha beta pruning on the Min-max game tree and also calculate the alpha and beta values for each non-pruned edge. Show pruned edges by a cross mark in the game tree.

[5][CO2]

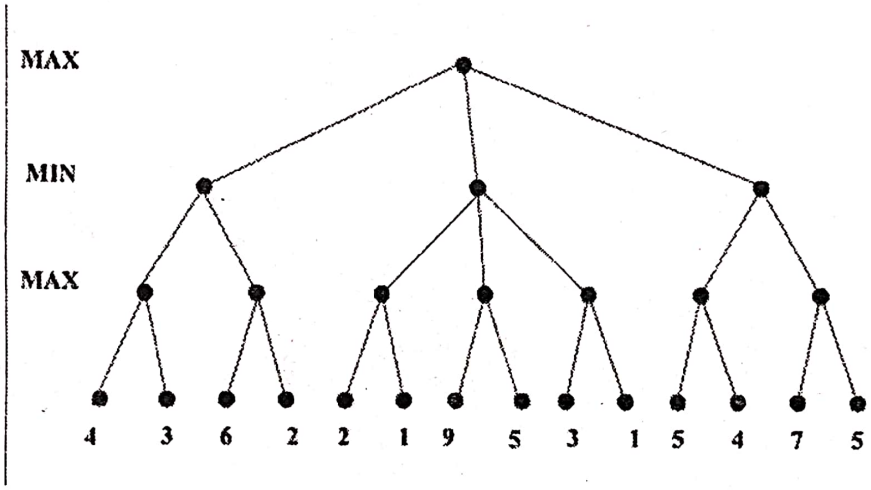

Figure 2

Q.4 Differentiate between the following:

i) Uniform Cost Search and Greedy Search

ii) A* and AO* algorithms

iii) Intelligent Agent and Intelligent Program

iv) Local maxima and Global maxima

v) Un-informed Search and Informed Search

[1.5x5=7.5][CO4]